

Teo fn suave soporte compacto denso lp

Teorema 1. $\mathcal{C}_c^\infty(\mathbb{R})$ es denso en $\mathcal{L}^p(\mathbb{R}, m)$ para todo $p \in [1, \infty)$.

Demostración: Por las demostraciones del Lem-aprox-indicatriz-continua-norma-lp/Lema 1 y del Teo-fn-continua-soporte-compacto-denso-lp/Teorema 1, basta con hallar $g \in \mathcal{C}_c^\infty(\mathbb{R})$ tal que $\mathbb{1}_{[a,b]}(x) \leq g(x) \leq \mathbb{1}_{[a-\varepsilon_1, b+\varepsilon_2]}(x)$ en función de $a, b, \varepsilon_1, \varepsilon_2$ dados. Para ello, basta con hallar $\varphi \in \mathcal{C}_c^\infty(\mathbb{R})$ tal que

$$\varphi(x) = \begin{cases} 0 & x \leq 0, \\ 1 & x \geq 1, \end{cases} \implies g(x) = \varphi\left(\frac{a-x}{\varepsilon_1}\right) \cdot \varphi\left(\frac{x-b}{\varepsilon_2}\right).$$

Para definir φ , usaremos la función ψ dada por

$$\psi(x) = \begin{cases} e^{-\frac{1}{x}} & x > 0, \\ 0 & x \leq 0. \end{cases} \implies \lim_{x \rightarrow 0^+} \forall n \in \mathbb{N} : \psi^{(n)}(x) = 0.$$

Sea $h(x) = \frac{1}{e} - \psi(x)$, entonces $\varphi(x) := e^e \cdot \psi(h(x))$ cumple lo pedido. ■

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